GGG-B-001222 (GSA-FSS)
October 11, 1967
SUPERSEDING DATA
(See 6.2)

INTERIM FEDERAL SPECIFICATION

BIT, SCREWDRIVER (AND ADAPTER, SCREWDRIVER BIT)

This Interim Federal Specification was developed by the Standardization Division, Federal Supply Service, General Services Administration, Washington, D.C. 20406, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers the most generally used screwdriver bits used for assembly operations and in the repair and maintenance of vehicles, machinery, and other items.

1.1.1 Federal specification coverage. Federal specifications do not cover all varieties of the commodity indicated by the titles of the specifications, or which are commercially available, but include only those generally used by the Federal Government.

1.2 Classification.

1.2.1 Types and classes. The tools covered by this specification shall be of the following types, and classes as specified (see 6.1).

Type I. Female socket.
- Class 1. Drag link pattern.
- Class 2. Phillips.
- Class 3. Reed and Prince.
- Class 4. Flat tip.

Type II. Male shank.
- Class 1. Phillips.
- Class 2. Reed and Prince.
- Class 3. Clutch.
- Class 4. Flat tip.
- Class 5. Socket head set and cap screw.

Type III. Adapter, screwdriver bit.

2. APPLICABLE DOCUMENTS

2.1 The following specifications and standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Standards:

Fed. Std. No. 123 - Marking for Domestic Shipment (Civilian Agencies).
Fed. Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Washington, D.C., Atlanta, Chicago, Kansas City, Mo., Ft. Worth, Denver, San Francisco, Los Angeles, and Seattle, Wash.

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(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military specifications:

MIL-H-15424 - Hand Tools, Packaging of.

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids shall apply.

United States of America Standards Institute (USASI):

B46.1 - 1962 Surface Texture.

(Application for copies should be addressed to the United States of America Standards Institute, 70 East 54th Street, New York, N.Y. 10012).

3. REQUIREMENT

3.1 Illustrations. The illustrations shown herein are for the purpose of identification and are not to preclude the purchase of items which are otherwise in accordance with the requirements of this specification.

3.2 Material. The material used in the manufacture of the tools shall be steel.

3.3 Marking. Each tool shall be marked in a plain and permanent manner with the manufacturer's name or trademark of such known character that the source of manufacture may be readily determined.

3.4 Hardness. Unless otherwise specified herein, the tools shall be hardened to Rockwell "C" 38 to 54 or equivalent (see 4.4.1).

3.5 Test loads. When tested in accordance with 4.4.3, tools shall withstand the test loads specified in the applicable tables without injury or permanent deformation which might affect the durability or serviceability of the tools and in no case shall visible distortion occur.

3.6 Finishes.

3.6.1 Surfaces. Surfaces shall be free from pits, nodules, burrs, and other defects which will adversely affect performance. Surfaces usually ground or otherwise finished and provided with a coating of chromium shall have a maximum surface roughness of 63 microinches. Other surfaces shall have a maximum roughness of 125 microinches (see 4.4.2).

3.6.2 Coatings. The coatings shall be adherent, smooth, continuous and free from pits, blisters, nodules, and any other defects which would interfere with their protective value and serviceability. The minimum thickness of the coating shall be as specified in 3.6.2.1 through 3.6.2.4 on all external visible surfaces which can be touched by a ball 0.750 inch in diameter. The coatings shall be one or a combination of the coatings specified herein.

3.6.2.1 Chromium plate. The coating shall be electrodeposited metals consisting of nickel, followed by chromium, the minimum thickness shall be 0.0002 inch for nickel and 0.00001 inch for chromium.

3.6.2.2 Nickel plate. The coating shall be electrodeposited nickel not less than 0.0002 inch thick.

3.6.2.3 Cadmium plate. The coating shall be electrodeposited cadmium not less than 0.0003 inch thick, and shall be subjected to a chemical or electrochemical chromating.

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3.6.2.4 Zinc plate. The coating shall be electrodeposited zinc not less than 0.0003 inch thick, and shall be subjected to a chemical or electrochemical chromating.

3.6.2.5 Hot phosphating. The coating shall consist of a chemically produced phosphate, followed with a coating of preservative oil.

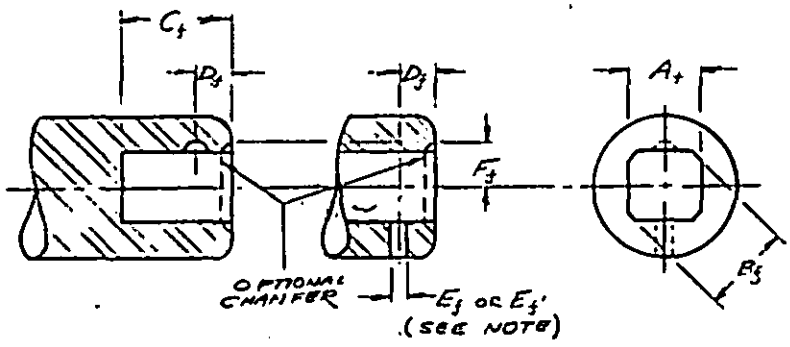
3.6.2.6 Oxide coating. The coating shall consist of a chemically produced oxide, followed with a coating of preservative oil.

3.7 Type I, female socket. Type I screwdriver bits shall have a square female socket on one end and a screwdriver point at the other end. Except for the class 1 drag link pattern, the bits shall be of either one integral unit or two unit design consisting of a body and screwdriver point. Means shall be provided in the two unit design for holding the point in the body in such a manner as to preclude the possibility of the point becoming loose and falling out under normal working conditions.

3.7.1 Female drive opening. All female drive openings shall be broached or punched in a smooth and well-defined manner. Female drive openings shall conform to figure 1 for the drive size specified, and be firmly attachable to corresponding size male drive tangs by the following methods:

3.7.1.1 On 3/8 inch and 1/2 inch drive openings. One or more faces of the female drive opening shall be drilled or recessed to engage the spring loaded steel ball or steel plunger on the corresponding male drive. If all the faces of the female opening are not recessed, the sockets shall be marked indicating the faces of the opening which are recessed.

3.7.1.2 On 1/4 inch drive openings. Faces of the female drive opening may or may not be drilled or recessed at the option of the manufacturer.



Drive size	Af		Bf	Cf	Df		Ef	Ef[1]	Ff
	Max.	Min.	Min.	Min.	Max.	Min.	Min.	Min.	Min.
Inch	Inches	Inches	Inches	Inches	Inch	Inch	Inch	Inch	Inch
1/4	0.260	0.253	0.335	0.312	0.159	0.147	0.090	0.120	0.145
3/8	.385	.378	.505	.438	.221	.209	.170	.232	.210
1/2	.511	.503	.670	.625	.315	.303	.201	.263	.275

[1] Ef dimensions shall be used in lieu of Ef dimensions when an internal chamfer is not provided.

FIGURE 1. Female socket standards.

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3.7.2 Class 1, drag link pattern. Class 1 bits shall have a tip thickness that is uniform to within .003 inch. The tip of the blade shall be at right angles tc the longitudinal axis of the blade within 2° across the tip width and thickness. The included angle between the bearing surfaces, from the tip of the blade to a line one-third of the tip width distant from the tip, shall not exceed 10°. The bit shall be through hardened to Rockwell "C" 48 minimum or equivalent and the reading taken within 1/4 inch of the tip end. Class 1 bits shall conform to table I for the drive and size number specified (see 6.1) and be similar to figure 2.

TABLE I. Type I, class 1

Drive size	Width ± 1/32	Thickness +0.010 - .005	Length overall maximum	Test load minimum
Inch	Inch	Inch	Inch	Inch-pounds
1/4	1/4	0.035	1	200
3/8	11/16	.116	1-1/4	950
1/2	47/64	.109	1-3/4	1500
1/2	15/16	.150	1-3/4	1650
1/2	1-3/16	.150	1-3/4	2000

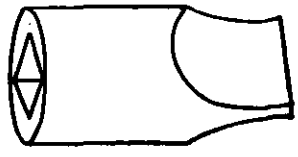


FIGURE 2. Type I, class 1.

3.7.3 Class 2, Phillips. Class 2 bits shall be through hardened to Rockwell "C" 54 minimum, or equivalent, and the reading taken as near the tip as practicle. Class 2 bits shall conform to table II, for the point and drive size specified (see 6.1), and be similar to figure 3. Tip dimensional conformance shall be determined with a gage conforming to figure 4. The tip of the blade flutes shall fall within the "C" dimension.

TABLE II. Type I, class 2

Point size	Drive size	Test load minimum	Overall length ±3/4	Nominal body diameter
Inch	Inch	Inch-pounds	Inches	Inch
1	1/4	25	2	3/16
2	1/4	60	2	1/4
3	3/8	150	3	5/16
3	1/2	150	3-1/2	5/16
4	3/8	200	3	3/8
4	1/2	200	3-1/2	3/8

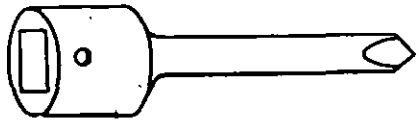
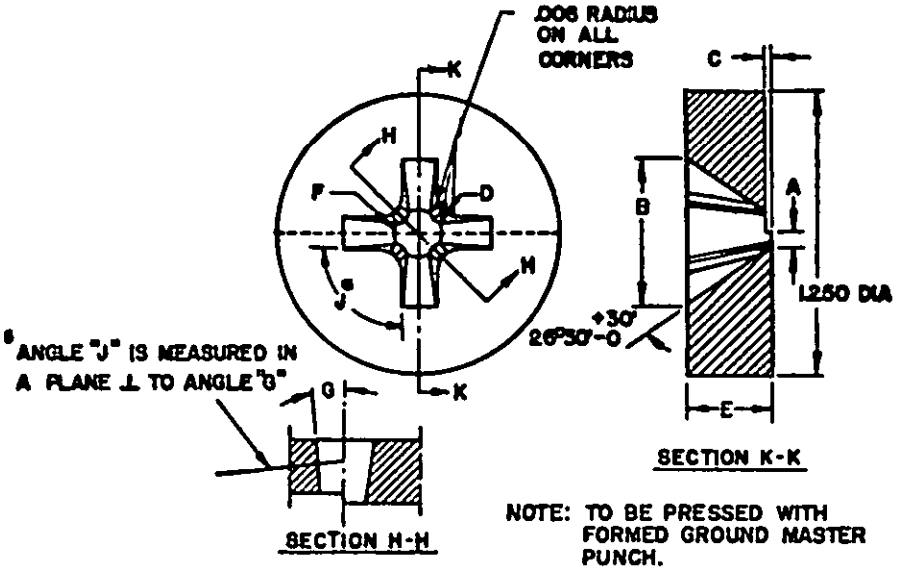


FIGURE 3. Type I, class 2.

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Point size	A ±0.0002	B (ref.)	C ±0.001	D (ref.)	E +0 -1/64	F ±0.0002	G +15' -0	J
Number	Inch	Inch	Inch	Inch	Inch	Inch		Degrees
1	0.0255	0.1765	0.010	0.0634	1/8	0.0194	7°	92
2	.0455	.2800	.010	.0971	3/16	.0303	5°45'	92
3	.0755	.4340	.010	.1538	9/32	.0496	5°45'	92
4	.1005	.5471	.010	.2003	11/32	.0711	7°	92

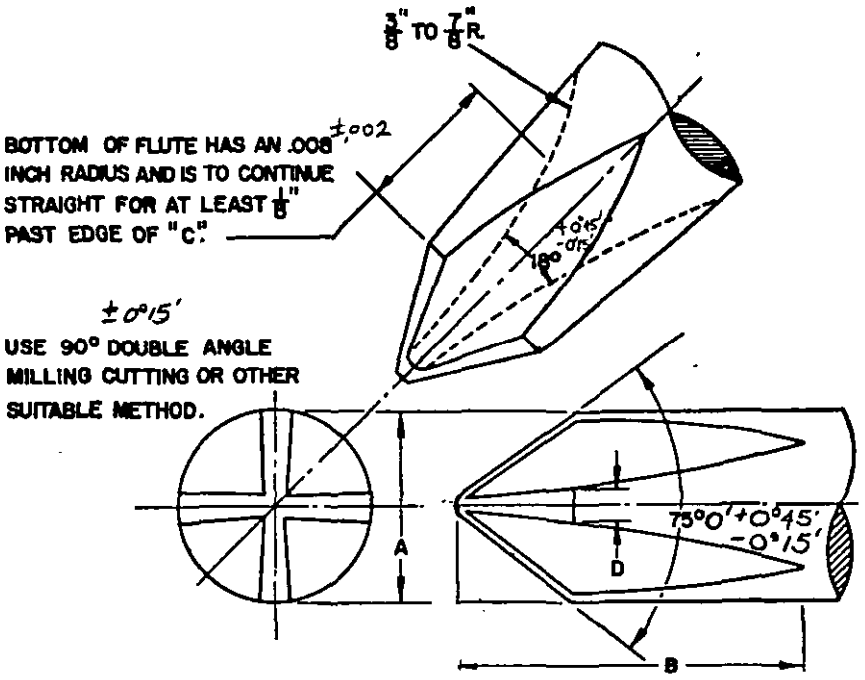
FIGURE 4. Gage for inspection Phillips screwdriver bit points.

3.7.4 Class 3, Reed and Prince. Class 3 screwdriver bit attachments shall be through hardened to Rockwell "C" 54 minimum or equivalent, and the reading taken as near the tip as practicle. Class 3 bits shall conform to table III, and figure 5 for the point and drive size specified (see 6.1), and be similar to figure 6.

TABLE III. Type I, class 3

Drive size	Point size A	Test load minimum	Overall length maximum
Inch	Inch	Inch-pounds	Inches
1/4	1/4	90	2-1/2
3/8	1/4	90	3
3/8	5/16	150	3
3/8	3/8	200	3
1/2	3/8	200	3-1/2

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Nominal size	A +.000 -.003	B +1/32	D +.001 -.002
1/4	.250	9/16	.049
5/16	.312	5/8	.059
3/8	.375	11/16	.068

FIGURE 5. Type I, class 3, and Type II, class 2.

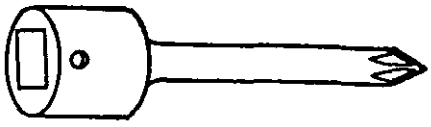


FIGURE 6. Type I, class 3.

3.7.5 Class 4, flat tip. Class 4 screwdriver bits shall be through hardened to Rockwell "C" 48 minimum and the reading taken within 1/4 inch of the tip end. All four sides and the end of the blade shall be symmetrical within 2°. The included angle between the bearing surfaces shall not exceed 10°. The extreme tip of the blade shall be at right angles to the longitudinal axis of the blade within 2° across the tip width and thickness. Class 4 bits shall conform to table IV for the drive size and bit specified (see 6.1), and be similar to figure 7.

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TABLE IV. Type I, class 4

Drive size	Bit		Overall length maximum	Test load minimum
	Width ±1/32	Thickness ±0.004		
Inch	Inch	Inch	Inches	Inch-pounds
1/4	3/16	0.032	3	25
1/4	1/4	.032	3	60
1/4	3/8	.042	3	150
3/8	1/4	.037	3	60
3/8	3/8	.062	3	200
1/2	9/16	.068	3-1/2	250

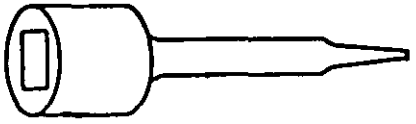
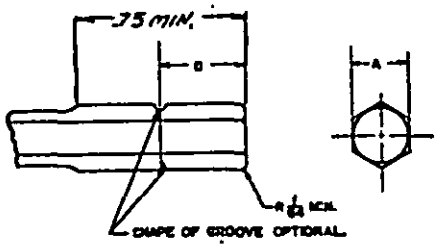


FIGURE 7. Type I, class 4.

3.8 Type II, male shank. Type II bits shall have a hexagon shaped shank conforming to figure 8 for the shank size specified (see 6.1). The bits shall be through hardened to Rockwell "C" 50 minimum or an equivalent case hardness .003 inch minimum deep.



Hexagon shank size A +0.000 - .003	Location of retaining groove B ±0.010
Inch	Inch
1/4	0.319
5/16	.399

FIGURE 8. Male shank.

3.8.1 Class 1, Phillips. Class 1 bits shall comply with the dimensional requirements of 3.7.3 as applicable. In addition, the bits shall comply with table V for the point and drive size specified (see 6.1) and be similar to figure 9.

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TABLE V. Type II, class 1

Shank size	Point size	Test load maximum	Overall length maximum
Inch	Inch	Inch-pounds	Inch
1/4	1	33	1
1/4	2	90	1
1/4	3	325	1
5/16	2	90	1-1/4
5/16	3	325	1-1/4
5/16	4	500	1-1/4

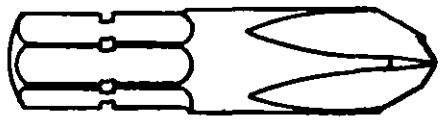


FIGURE 9. Type II, class 1.

3.8.2 Class 2, Reed and Prince. Class 2 bits shall conform to the dimensional requirements of 3.7.4 and table VI for the point and drive size specified (see 6.1), and be similar to figure 10.

TABLE VI. Type II, class 2

Shank size	Point size	Test load minimum	Overall length minimum
Inch	Inch	Inch-pound	Inches
1/4	1/4	90	1
5/16	5/16	325	1-1/4

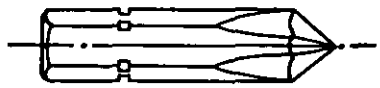


FIGURE 10. Type II, class 2.

3.8.3 Class 3, clutch. Class 3 bits shall comply with table VII and figure 11 for the point and drive size specified (see 6.1).

TABLE VII. Type II, class 3

Shank size	Nominal clutch size	Dimension		Radius		Overall length minimum	Test load minimum
		B +0.003 - .002	D minimum	F ± 0.0005	G -0.0020		
Inch		Inch	Inch	Inch	Inch	Inches	Inch-pounds
5/16	5/32	0.056	3/16	0.0780	0.0515	1-5/16	60
5/16	1/4	.090	9/32	.1210	.0788	1-5/16	150
5/16	5/16	.112	3/8	.1510	.0995	1-7/16	500

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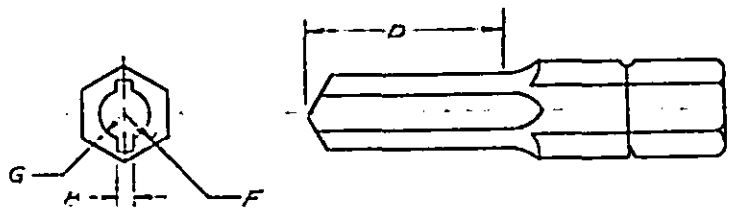


FIGURE 11. Type II, class 3.

3.8.4 Class 4, flat tip. Class 4 screwdriver bits shall be of the flat tip pattern. The bits shall conform to 3.7.5 and table VIII for the shank size and blade specified (see 6.1) and be similar to figure 12.

TABLE VIII. Type II, class 4

Shank size	Blade		Test load minimum	Overall length maximum
	Thickness +0.000 - .003	Width +0.010 - .005		
Inch	Inch	Inch	Inch-pounds	Inches
1/4	0.034	3/16	35	1-1/2
1/4	.040	7/32	75	1-1/2
1/4	.042	1/4	90	1-1/2
1/4	.046	9/32	135	1-1/2
1/4	.050	19/64	180	1-1/2
1/4	.055	11/32	230	1-1/2
5/16	.055	3/8	275	1-1/2

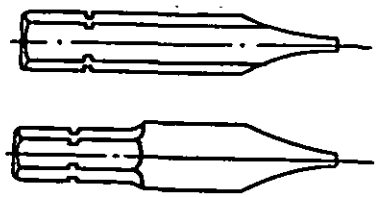


FIGURE 12. Type II, class 4.

3.8.5 Class 5, socket head set and cap screw. Class 5 bits shall be suitable for internal wrenching. The point shall be through hardened to Rockwell "C" 50 to 64, or equivalent, and the reading taken within 1/4 inch of the tip end. The bit shall conform to table IX for the drive and point size specified (see 6.1) and be similar to figure 13.

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TABLE IX. Type II, class 5

Shank size	Nominal point size	Distance across flats		Overall length maximum	Test load minimum
		Minimum	Maximum		
Inch		Inch	Inch	Inches	Inch-pounds
1/4	1/16	0.0615	0.0625	1	4
1/4	5/64	.0771	.0781	1	8
1/4	3/32	.0927	.0938	1	12
1/4	7/64	.1078	.1093	1	15
1/4	1/8	.1235	.1250	1	20
1/4	9/64	.1391	.1406	1	25
1/4	5/32	.1547	.1562	1-5/16	35
1/4	3/16	.1860	.1875	1-5/16	45
1/4	7/32	.2175	.2188	1-5/16	100
1/4	1/4	.2485	.2500	1-5/16	150
5/16	5/16	.3110	.3125	1-5/16	275
1/4	3/8	.3725	.3750	1-1/4	175
5/16	3/8	.3735	.3750	1-1/4	500



FIGURE 13. Type II, class 5.

3.9 Type III, adapter, screwdriver bit. Screwdriver bit adapters shall have at one end a square female socket conforming to 3.7.1 for attachment to a driving medium. The other end shall have a hexagonal female socket to receive type II bits.

3.9.1 Bit locking mechanism. A spring loaded bit locking mechanism or other suitable locking device shall be incorporated in the hexagon socket which shall securely retain the bit in the holder and permit its removal without the aid of any tools or keys. Type III adapters shall conform to table X for the drive and hexagon size specified, and be similar to figure 14.

TABLE X. Type III

Nominal square drive size	Hexagon bit opening		Hexagon depth +1/32 -0	Location of bit retainer from front end +0.002 -0.030	Overall length maximum	Test load minimum
	Distance across flats					
	Minimum	Maximum				
	Inch	Inch	Inch	Inch	Inches	Inch-pounds
1/4	0.252	0.257	3/8	0.056	1	350
3/8	.252	.257	3/8	.056	1-1/4	350
3/8	.316	.322	31/64	.078	1-1/2	500
1/2	.316	.322	31/64	.078	1-1/2	500



FIGURE 14. Type III.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection of materials and components. In accordance with 4.1 above, the supplier is responsible for insuring that materials and components used were manufactured, tested and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification.

4.2 Sampling procedures. Sampling procedures shall be in accordance with MIL-STD-105. Data for sampling shall be as indicated in table XI.

TABLE XI. Sampling Data

Category	Sample Unit	Inspection Level	Acceptable Quality Level	AQL Expressed In Terms Of	Reference
Visual examination	1 ea.	II	Major Minor 2.5 6.5	Defects per hundred units	4.3.1
Dimensional examination	1 ea.	S-4	1.5	Defects per hundred units	4.3.2
Testing	1 ea.	S-3	1.0	Percent defective	4.4
Examination of preparation for delivery	one container	S-2	4.0	Defects per hundred units	4.5

4.3 Examination.

4.3.1 Visual examination. Each sample unit shall be examined for any nonconformance in design, material, finish, coating, construction, workmanship, and marking. Defects are listed in table XII.

TABLE XII. Classification of defects

Examine	Defect	Major	Minor
Finish	Surfaces not finished as specified	X	
	Coatings (as applicable) not adherent, smooth, continuous or free from uncoated areas, pits, blisters, nodules, and other defects affecting protective value		X
Design	Type, class, style, and size not as specified	X	
Material	Material not as specified	X	
Construction and workmanship	Broaching not smooth or well defined	X	
	Female opening not drilled nor recessed	X	

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4.3.2 Dimensional examination. Each sample unit shall be examined for any nonconformance with dimensional requirements.

4.4 Testing. Each sample unit shall be tested in accordance with tests under 4.4.1, 4.4.2, 4.4.3, and 4.4.4 as applicable.

4.4.1 Hardness. Hardness determination shall be made in accordance with Method 243 of Federal Standard 151.

4.4.2 Finish. Finish determination shall be made in accordance with United States of America Standards Institute Bulletin-B46.1-1962.

4.4.3 Test load.

4.4.3.1 Mandrels. Mandrels for torsion tests shall be of sufficient strength and hardness to resist fracture or deformation when transmitting the required test load.

4.4.3.1.1 Flat tip bits. The mandrels for flat tip bits shall be slotted to a depth not greater than the thickness of the blade plus the respective positive tolerance permitted. The slot width shall be at least as great as the width of the bit being tested.

4.4.3.1.2 Phillips. The recess in the mandrel shall conform to the applicable dimensions of figure 4.

4.4.3.1.3 Reed and Prince. The recess in the mandrel shall be of such configuration as to receive a point of the maximum size shown in figure 5 and be of such depth that the wing spread at the surface will be as follows:

Blade size	Wing-spread maximum
Inch	Inch
.250	.252
.312	.315
.375	.377

4.4.3.1.4 Clutch and socket head. The recess in the fixture for clutch and socket-head bits shall conform to the dimensions of the recess in the applicable commercial type screw head for which the bit is designed to drive.

4.4.3.2 Application of test load. The load shall be applied with suitable torque-producing machines or torque wrenches for a period of 10 seconds and then removed.

4.5 Inspection of preparation for delivery requirements. An inspection shall be made to determine that the preservation, packaging, packing, and marking comply with the requirements of section 5 of this specification. Defects shall be scored in accordance with table XIII. The sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot.

TABLE XIII. Classification of Defects

Examine	Defects
Markings (exterior and interior)	Omitted; incorrect; illegible; improper size, location, sequence, or method of application.
Materials	Any components missing, or damaged.
Workmanship	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Distortion of container.

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5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, and packing. Unless otherwise specified (see 6.1), the bits shall be preserved, packaged, and packed in accordance with MIL-H-15424. The level of preservation, and packaging shall be A or C and the level of packing, A, B, or C as specified (see 6.1).

5.1.1 As part of the requirements of section of this specification, the following is considered as part of table I of MIL-H-15424:

COLUMN 1	COLUMN 2	COLUMN 3	COLUMN 4	COLUMN 6
Bit, screwdriver and adapter screwdriver bit				
Female socket	P-8, P-9	I	1*	140
Male shank	P-8, P-9	I	1*	140
Adapter, screwdriver bit	P-8, P-9	I	1*	140

*Preserved and packaged in a polyethylene bag. The bag shall be closed or sealed.

5.2 Marking.

5.2.1 Civil agencies. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military activities. In addition to markings required by the contract or order, the interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in procurement documents:

(a) Title, number and date of this specification.
(b) Type, and class required (see 1.2.1).
(c) Size required (see applicable table).
(d) Level of preservation, packaging, and packing required (see 5.1).
(e) Additional marking, if required (see 5.2).

6.2 When finalized, this specification is intended to supersede, in part, GGG-W-641d Wrench, Socket; (and Socket, Handles, and Attachments for Socket Wrenches; Hand).

6.2.1 Types, classes and styles. The following list indicated the types, and classes in this specification and the items transferred from other specifications.

GGG-B-00122	GGG-W-641d
Type I	Type IV
Class 1	Class 6
Class 2	Style A
Class 3	Form 1
Class 4	Form 2
Type II	Form 3
Class 1	Form 4
Class 2	Style B
Class 3	Form 1
Class 4	Form 2
Class 5	Form 3
Type III	Form 4
	Form 5
	Class 7

6.3 For civil agency procurements. When level B packaging is required for civil agency procurements, the level A requirements in 5.1.1 shall apply.

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